Amritaraj Nair

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Education

Texas A&M University — B.S. Computer Science Honors, Minor in Mathematics *Expected May 2029*
Relevant Coursework: Data Structures & Algorithms, C++ Design, Discrete Structures, Engineering Calculus III, Python Programming

Experience

AI Modeling for Livestock Systems — TAMU Research *Aug 2025 – Present*
- Applied **agent-based modeling (Mesa)** and machine learning to optimize livestock nutrition under economic and environmental constraints.
- Migrated legacy **R workflows to Python**, improving reproducibility, scalability, and cross-pipeline integration.
- Built cross-platform simulation systems in **Python, Java, and NetLogo** to evaluate sustainability and production outcomes.

Cybersecurity & Large Language Models — TAMU Research *Aug 2025 – Present*
- Evaluated **Cyber Reasoning Systems (CRS)** powered by LLMs, including Theori RoboDuck and TAMU AYNIAFB.
- Benchmarked vulnerability detection across real-world **C, Python, and Java** repositories.
- Analyzed model performance across heterogeneous systems to assess reliability and generalization.

Maroon Fund — Scholars of Finance *Feb 2025 – Present*
Quantitative Developer
- Developed **quantitative models and screening frameworks** for equity research across a **$70K+ student-managed fund**.
- Engineered **data pipelines and valuation tools** using financial datasets to improve investment analysis efficiency.
- Automated **research workflows**, delivering data-driven insights for stock pitch processes and portfolio decisions.

MagNet Agents — Cornell-Backed Legal Tech Startup *May 2025 – Aug 2025*
Software Engineering Intern
- Engineered **data pipelines and web scraping systems** to extract and structure high-value legal lead data from public sources, enabling automated client acquisition workflows.
- Developed **AI-powered outreach generation features** using LLMs to create personalized messaging and legal opportunity summaries, improving engagement and conversion.
- Built core **backend systems for lead matching and tracking**, connecting lawyer profiles with relevant prospects and supporting a platform generating **$50K+ ARR**.
- Collaborated in a fast-paced startup environment to ship **end-to-end product features**, contributing to a Cornell-backed platform at the intersection of legal tech and AI.

Projects

AlphaForge | *Founder & Solo Developer* *March 2026*
- Built **AlphaForge**, a full-stack quantitative investing platform enabling users to design, test, and deploy trading strategies.
- Developed a multi-level **Quant IDE** with a Scratch-like builder for beginners and an advanced coding environment supporting multi-language strategies.
- Implemented **backtesting and paper trading systems**, improving strategy validation and iteration efficiency.
- Engineered a real-time **market tracker** covering stocks, ETFs, indices, and crypto with integrated analytics and financial news.
- Designed an interactive **learning platform** to explain financial metrics directly within the user interface.
- Integrated **Yahoo Finance, Alpaca, and Seeking Alpha APIs**; deployed on **Vercel** for scalable, low-latency performance.

Shot Sensei | *Lead Developer & Co-Founder* *Mar 2026 – Present*
- Built a computer vision hackathon project for competitive and training-based pickleball, winning the **Hookem Hacks 2026 (Startup Ready Award)**.
- Developed a real-time gameplay mode where users play against an AI bot, alongside a training mode using **Gemini** and **ElevenLabs** to analyze and coach all pickleball strokes (serve, volley, forehand, backhand).
- Implemented pose and shot detection using **OpenCV** and **YOLOv8** for stroke classification and performance feedback.
- Engineered backend data infrastructure with **Supabase** to track player win/loss records and shot-level scoring analytics.
- Invited back to the **McCombs School of Business at UT Austin** to pitch Shot Sensei to **Pear VC (Khalil Fuller)**, presenting it as both a competitive AI-powered pickleball game playable anytime, anywhere and a personalized training system.

ClinicalHours | *Lead Engineer* *Dec 2025 – Present*
- Built an **AI-powered email and call automation system** using **Gmail API and GoHighLevel (GHL)** to act as a virtual receptionist, scheduling meetings with clinics and accelerating application workflows for pre-med students.
- Integrated **MapBox, Google APIs, and Resend API** to enable geolocation-based clinic discovery and automated communication pipelines.
- Processed large-scale **U.S. hospital datasets** in Python, transforming raw data into structured, production-ready formats.
- Implemented premium features like AI resume and application tools using **OpenAI and Gemini**. Partnered with clinics like **BCS Free Health Clinic**

Technical Skills & Awards

Languages: Python, C++, Java, TypeScript, SQL **Frameworks:** React, Flask, FastAPI, Pandas, NumPy, Matplotlib
Tools: Git, Docker, AWS EC2/IAM, Google Cloud, Supabase, CI/CD
Awards: UT Austin Hook'em Hacks 2026 **Winner** (Most Startup Ready + Multimodal Track) ·TidalTAMU Google Gemini Track **1st Place** (2026) · Outstanding Undergraduate Researcher Award· President's Endowed Scholar · Google Labs Makeathon · FRC 4192 Impact Award